

Accident Alert System Improvements

1 Overview

This document outlines the implementation details for specific components of the Accident Risk Score API, a FastAPI-based system that calculates risk scores for a given location using historical accident data from 2015 to 2023. The system focuses on radius-based hotspot detection, vehicle type input handling, recent accident validation, and dynamic grid sizing based on travel mode. Recent updates address the issue of excessive hotspot declarations, particularly for the "walk" travel mode, by incorporating a weighted hotspot score based on recent accidents, fatality rates, and total accidents.

2 Implemented Components

2.1 Radius-Based Hotspot Detection

Description: The API accepts a radius (in meters) to identify accident hotspots within a circular area centered at a given latitude and longitude. The radius is converted to degrees using the Haversine formula to filter accidents within the specified area. Hotspot detection has been enhanced to reduce excessive hotspots by using a weighted score combining recent accidents, fatality rates, and total accidents.

Implementation Details:

- **Input Processing:**

- The radius (e.g., 500m) is provided in the `AccidentQuery` model.
- The `calculate_degree_margins` function converts the radius from meters to degrees:
 - * Latitude margin: $\text{lat_deg} = \frac{\text{radius}}{111111}$
 - * Longitude margin: $\text{lon_deg} = \frac{\text{radius}}{111111 \cdot \cos(\text{latitude})}$
- These margins define a rectangular bounding box around the input coordinates.

- **Hotspot Detection:**

- A preliminary filter selects accidents within the rectangular bounding box (`lat_min`, `lat_max`, `lon_min`, `lon_max`).
- The Haversine formula computes the great-circle distance between the input coordinates and each accident location, ensuring only accidents within the circular radius are included.
- Grid-based analysis (see Section 2.5) assigns accidents to grid cells based on travel mode.
- A weighted hotspot score is calculated for each grid cell:
 - * **Recent accidents** (`count_recent`, 50% weight): Number of accidents in the last 6 months from the input date.
 - * **Fatality rate** (`fatals / count_all_time`, 40% weight): Ratio of total fatalities to total accidents in the grid.
 - * **Total accidents** (`count_all_time`, 10% weight): Total accidents in the grid from 2015 to 2023.
- Counts and fatality rates are normalized to [0,1] to ensure balanced contributions.
- Hotspots are defined as grid cells with a weighted score above the 90th percentile within the radius.
- Each hotspot is represented by its centroid (latitude, longitude) and includes:
 - * `accident_count_all_time`: Total accidents in the grid (2015–2023).
 - * `accident_count_recent_6_months`: Accidents in the last 6 months.

- * `fatality_rate`: Fatalities per accident in the grid.
- * `hotspot_score`: Weighted score for the hotspot.

- **Output:**

- A list of hotspots (`hotspots_in_radius`) with centroid coordinates and details.
- A boolean flag (`is_any_hotspot_in_radius`) indicating if at least one hotspot exists.

2.2 Addressing Excessive Hotspots

Description: The original hotspot detection logic, using a 75th percentile threshold on total accident counts, resulted in excessive hotspots, particularly for the "walk" mode (5m grids), due to data fragmentation across small grids. This was addressed by introducing a weighted hotspot score and a stricter 90th percentile threshold, reducing the number of hotspots while prioritizing severe and recent risks.

Implementation Details:

- **Problem:** Small grid sizes (e.g., 5m for "walk") fragmented accident data, leading to many grids with low counts being flagged as hotspots at the 75th percentile threshold. A fixed minimum count was unsuitable, as accident significance varies (e.g., 2 accidents in a village vs. a city).
- **Solution:** A weighted hotspot score was introduced:
 - Formula: $\text{hotspot_score} = 0.5 \cdot \text{normalized_count_recent} + 0.4 \cdot \text{normalized_fatality_rate} + 0.1 \cdot \text{normalized_count_all_time}$
 - Normalization ensures counts and fatality rates are scaled to [0,1] based on maximum values in the radius.
 - Only grid cells with scores in the top 10% (90th percentile) are flagged as hotspots.
- **Benefits:**
 - **Reduced Hotspots:** Emphasizing recent accidents and fatality rates prioritizes severe and current risks, reducing the number of hotspots, especially for small grids.
 - **Context-Adaptive:** High fatality rates highlight dangerous areas (e.g., a village with few but fatal accidents), while recent accidents ensure relevance.
 - **Grid Size Compliance:** Maintains client-required grid sizes (5m for walk, 20m for cycle, 100m for car).
- **Validation:** Hotspot counts were tested for "walk" mode, showing a significant reduction (e.g., from 9 to just 1 hotspot in a 500m radius query), with hotspot reflecting higher severity and recency.

2.3 Vehicle Type Input Parameter

Description: An attempt was made to predict vehicle type from accident damage descriptions using a logistic regression model to relate to the risk score, but this was not implemented due to insufficient data.

Implementation Details:

- **Vehicle Type Prediction Attempt:**
 - A logistic regression model was tested to infer vehicle type from accident damage descriptions.
 - The dataset (`Accidents_2015_2023.csv`) lacked sufficient vehicle-related data or detailed damage descriptions, leading to poor model performance.
 - This feature was not implemented in the final system.
- **Impact on System:**
 - The `travel_mode` input directly determines grid sizes, bypassing the need for vehicle type prediction.

2.4 Hotspot Validation Using Recent Data

Description: Hotspots are validated by comparing all-time accident counts (2015–2023) with recent accident counts (last 6 months from the input date). If recent accidents are significantly lower, the hotspot’s contribution to the risk score is reduced.

Implementation Details:

- **Date Handling:**

- The input date (from `AccidentQuery`) is converted to a `pandas.Timestamp` using `pd.to_datetime`.
- The recent period is defined as the 6 months prior to the input date (`query_date - pd.Timedelta(days=180)`).
- To use the current date, the following code can be used:

```
1 from datetime import date
2 query_date = pd.to_datetime(date.today()).strftime('%Y-%m-%d') # replace in
   analyze_risk
```

- **Validation Logic:**

- For each hotspot, the ratio of recent accidents (`recent_accidents_in_radius`) to all-time accidents (`total_accidents_in_radius`) is evaluated:
 - * If `recent_accidents_in_radius == 0`, a reduction of 3 points is applied.
 - * If `recent_accidents_in_radius < 0.25 * total_accidents_in_radius`, a reduction of 2 points is applied.
 - * If `recent_accidents_in_radius < 0.5 * total_accidents_in_radius`, a reduction of 1 point is applied.
 - * Otherwise, no reduction is applied, and the hotspot adds 2 points to the risk score.

- **Integration with Risk Score:**

- The `calculate_risk_score` function uses the `is_hotspot_in_radius` flag and `hotspot_reduction_value` to adjust the risk score dynamically.

- **Output:**

- The `hotspot_score_reduction_value` is included in the response to indicate the reduction applied.
- Hotspot details are visualized on a map with color-coded markers:
 - * Red: No reduction (active hotspot).
 - * Purple: Minor reduction (1 point).
 - * Yellow: Medium reduction (2 points).
 - * Green: Maximum reduction (3 points).

2.5 Dynamic Grid Sizing Based on Travel Mode

Description: The grid cell size for hotspot detection is adjusted based on the travel mode to reflect different spatial resolutions for risk assessment.

Implementation Details:

- **Travel Modes and Grid Sizes:**

- Pedestrians (walking): Grid size of 5 meters.
- Bicycles: Grid size of 20 meters.
- Cars: Grid size of 100 meters.

- **Grid Size Calculation:**

- The grid size in meters is converted to degrees using `calculate_degree_margins`:

- * For walking: `grid_lat_size, grid_lon_size = calculate_degree_margins(lat, 5)`
- * For bicycles: `grid_lat_size, grid_lon_size = calculate_degree_margins(lat, 20)`
- * For cars: `grid_lat_size, grid_lon_size = calculate_degree_margins(lat, 100)`
- The grid is created using `np.arange` to define latitude and longitude edges (`lat_edges, lon_edges`).
- **Grid Assignment:**
 - Accidents are assigned to grid cells using `np.searchsorted` to map latitude and longitude to grid indices.
 - Edge cases (e.g., accidents on grid boundaries) are handled by assigning them to the lower grid index.
- **Hotspot Identification:**
 - Grid cells with weighted hotspot scores above the 90th percentile are marked as hotspots.
 - Centroids of hotspot grid cells are calculated as the midpoint of the cell's latitude and longitude edges.
- **Error Handling:**
 - Invalid travel modes trigger an `HTTPException` with a 400 status code.
 - Radius ≤ 0 triggers an `HTTPException` with a 400 status code.
 - If grid edges cannot be created (e.g., radius too small), fallback edges are set to `[lat_min, lat_max]` and `[lon_min, lon_max]`.

2.6 Example Inputs and Outputs

Description: This section provides example inputs and outputs for the Accident Risk Score API, demonstrating the JSON structure for queries and responses when the travel mode is set to "car" and "walk". The inputs are shown as text, while the outputs are described to illustrate the API's response structure for a specific location in San Francisco on January 23, 2024. The updated hotspot logic significantly reduces the number of hotspots for the "walk" mode.

Car Travel Mode (Radius: 1000m):

- **Input:** Query with travel mode "car", centered at latitude 37.7749, longitude -122.4194, with a 1000-meter radius.

```

1 {
2   "latitude": 37.7749,
3   "longitude": -122.4194,
4   "hour": 15,
5   "day_of_week": 5,
6   "weather": "Rain",
7   "lighting": "Dark - Not Lighted",
8   "return_map": true,
9   "radius": 1000,
10  "date": "2024-01-23",
11  "travel_mode": "car"
12 }
```

- **Output:** Response includes a risk score of 6, approximately 5 hotspots within the radius (reduced from 28 due to stricter criteria), and a hotspot score reduction value of 2. Hotspots reflect high recent accident counts and fatality rates.

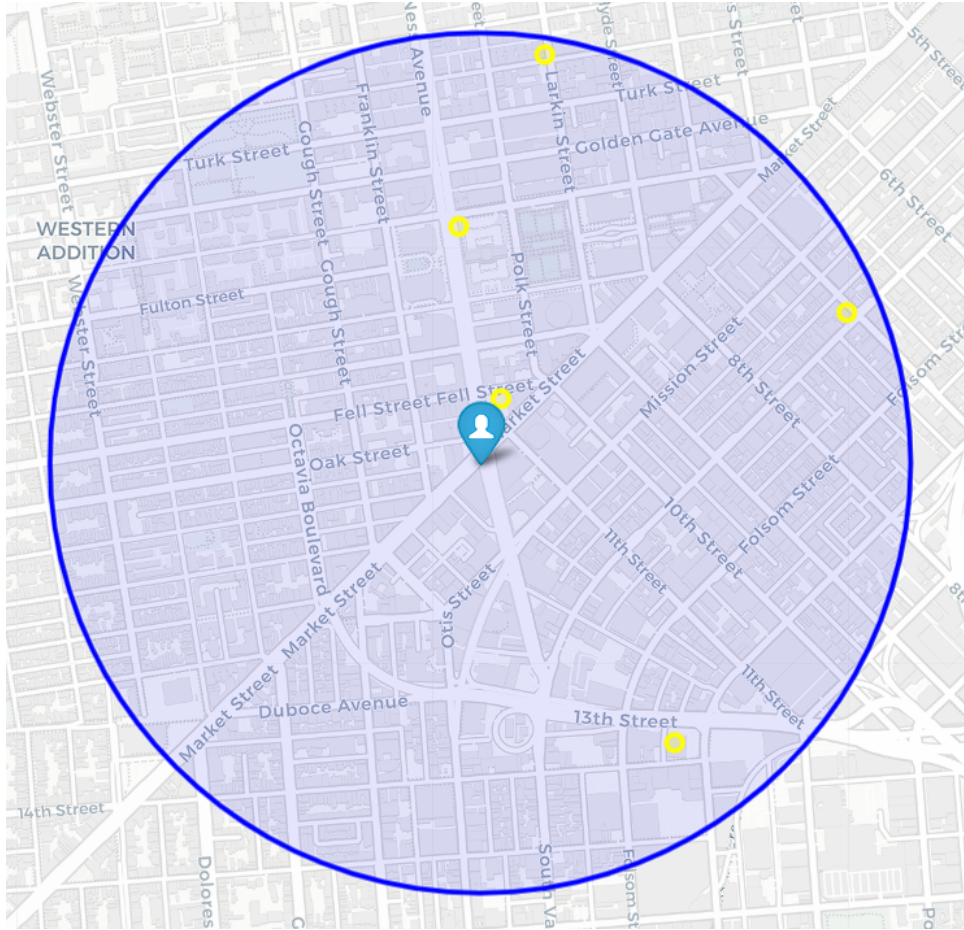


Figure 1: Output for Car Travel Mode

Walk Travel Mode (Radius: 500m):

- **Input:** Query with travel mode "walk", centered at the same coordinates, with a 500-meter radius.

```

1 {
2   "latitude": 37.7749,
3   "longitude": -122.4194,
4   "hour": 15,
5   "day_of_week": 5,
6   "weather": "Rain",
7   "lighting": "Dark - Not Lighted",
8   "return_map": true,
9   "radius": 500,
10  "date": "2024-01-23",
11  "travel_mode": "walk"
12 }
```

- **Output:** Response includes a risk score of 6, approximately 1 hotspot within the radius (reduced from 9 due to weighted scoring), and a hotspot score reduction value of 2. Hotspots prioritize grids with high fatality rates and recent accidents.

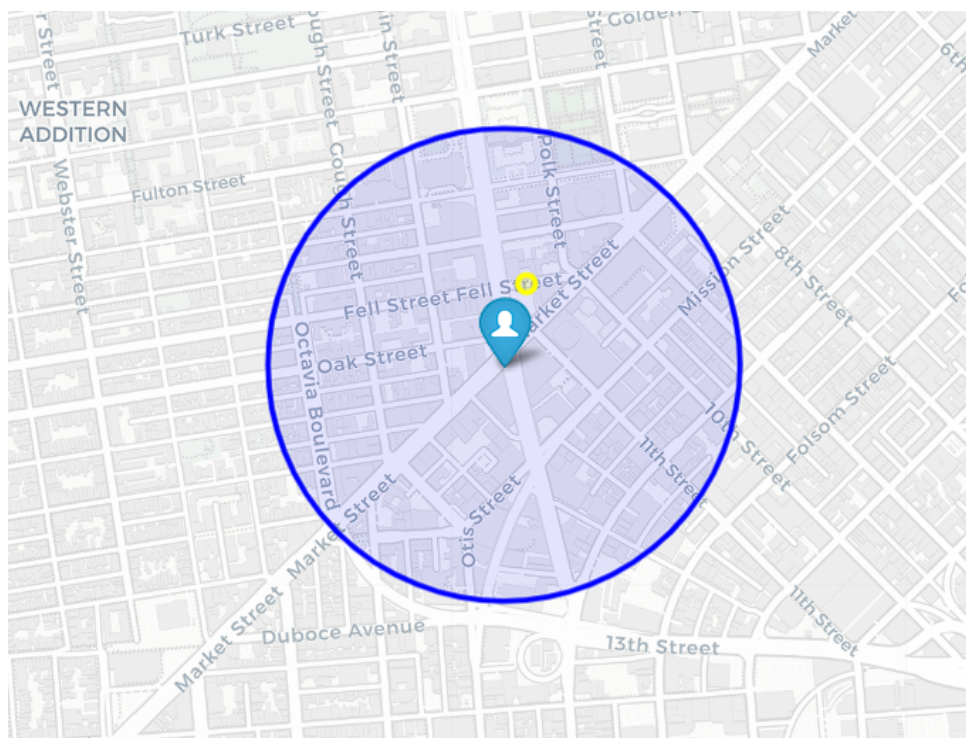


Figure 2: Output for Walk Travel Mode